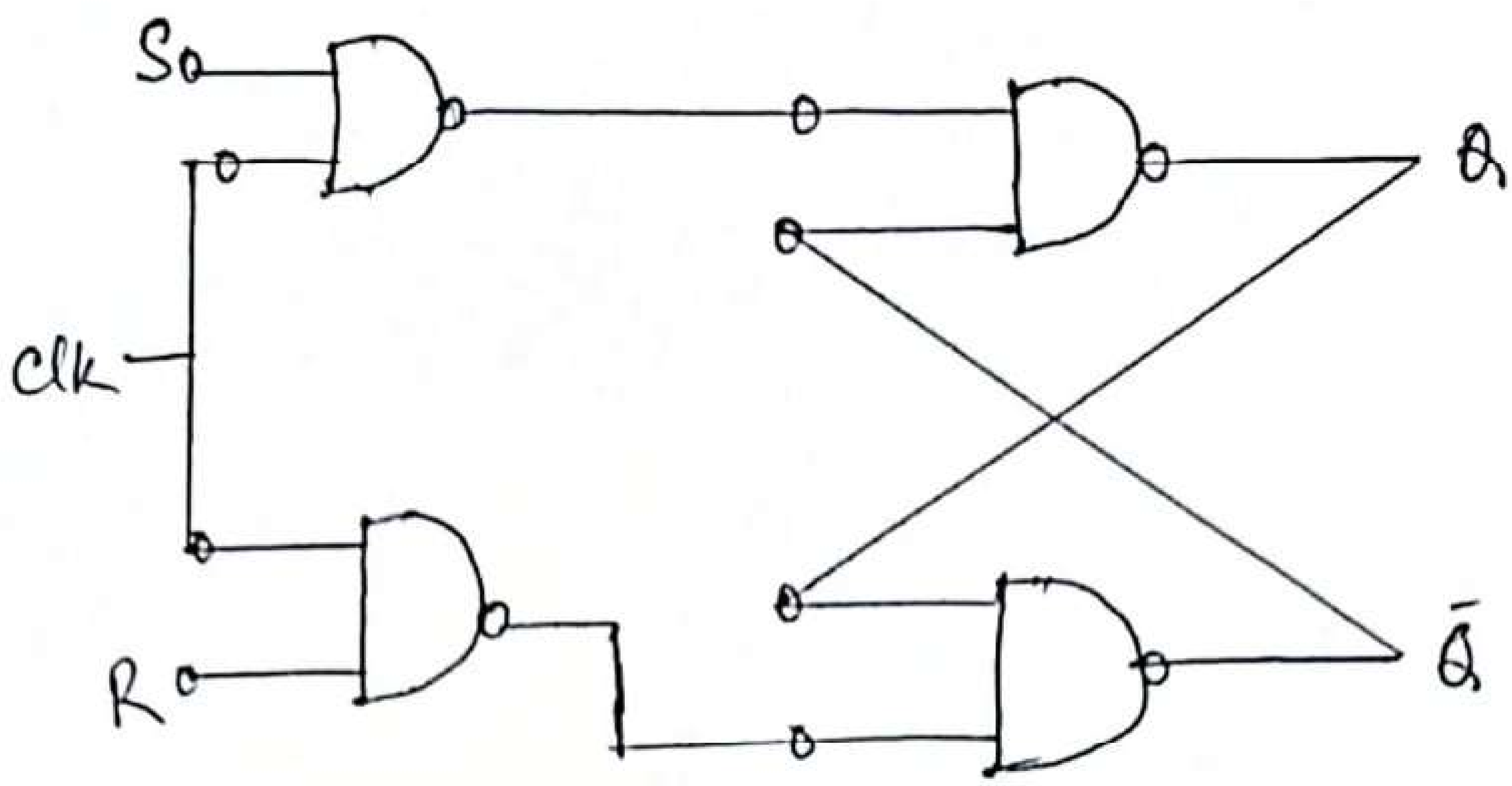
6.1

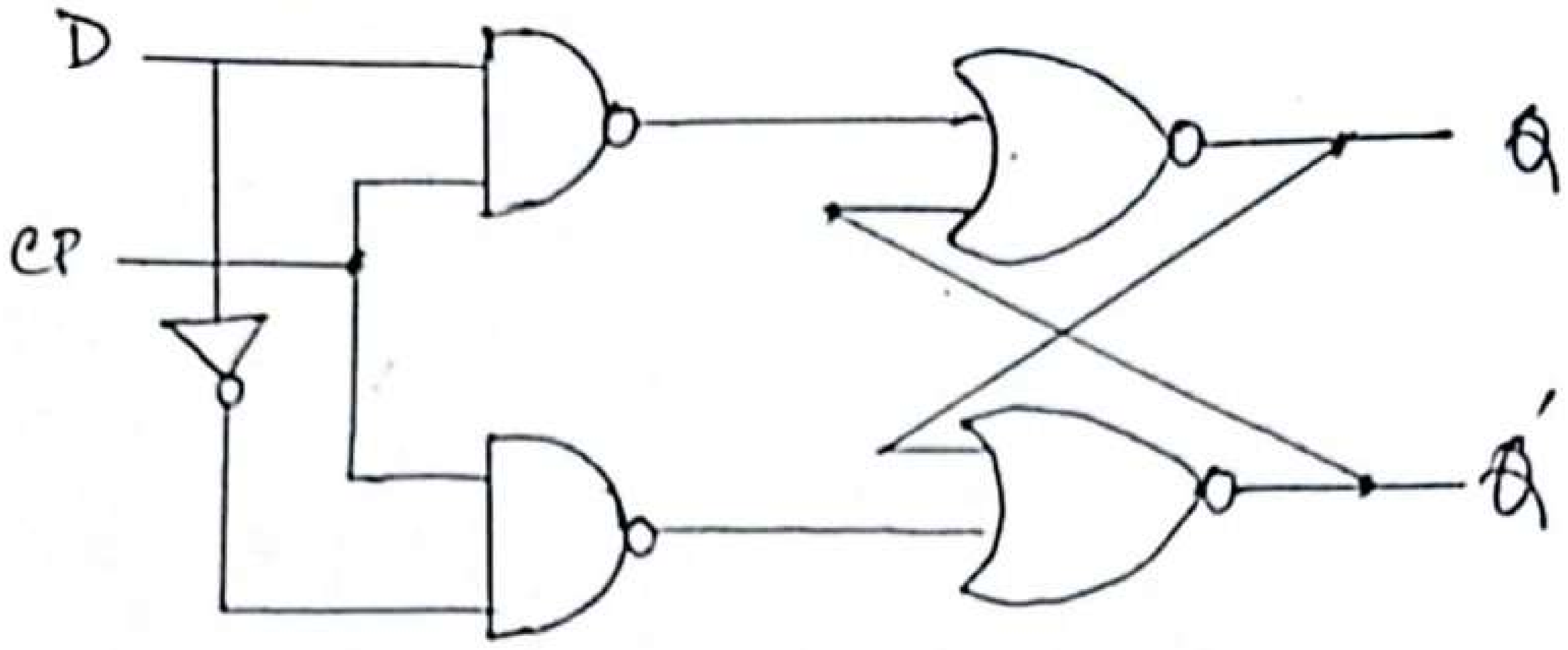


Truth table:

Clk	S	R	Q	$\bar{Q}$
0	X	X	Q_{prev}	$\bar{Q}_{prev}$
1	0	1	0	1
1	1	0	1	0
1	0	0	Q_{prev}	$\bar{Q}_{prev}$
1	1	1	Not allowed	Not allowed

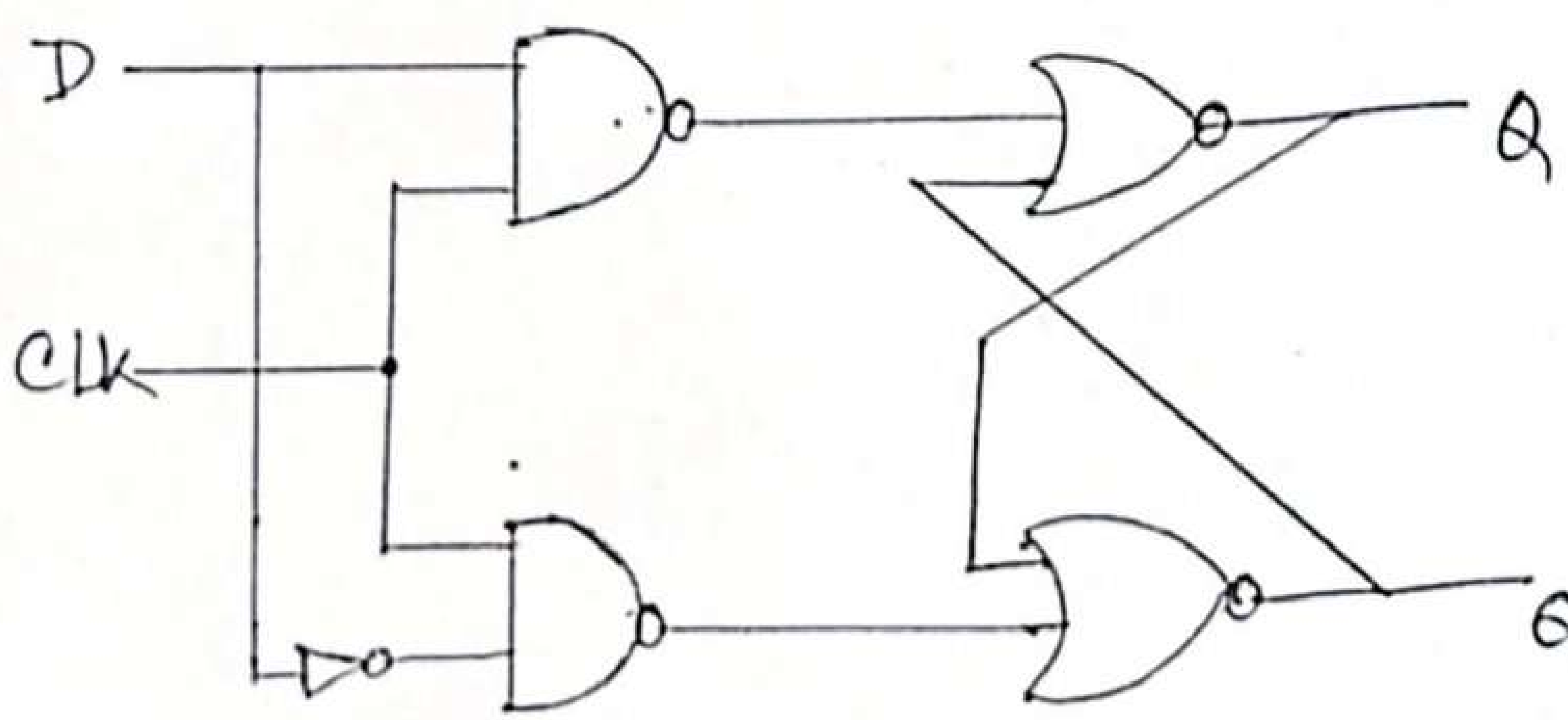
6.2

D. flip-flop with AND & NOR gates.



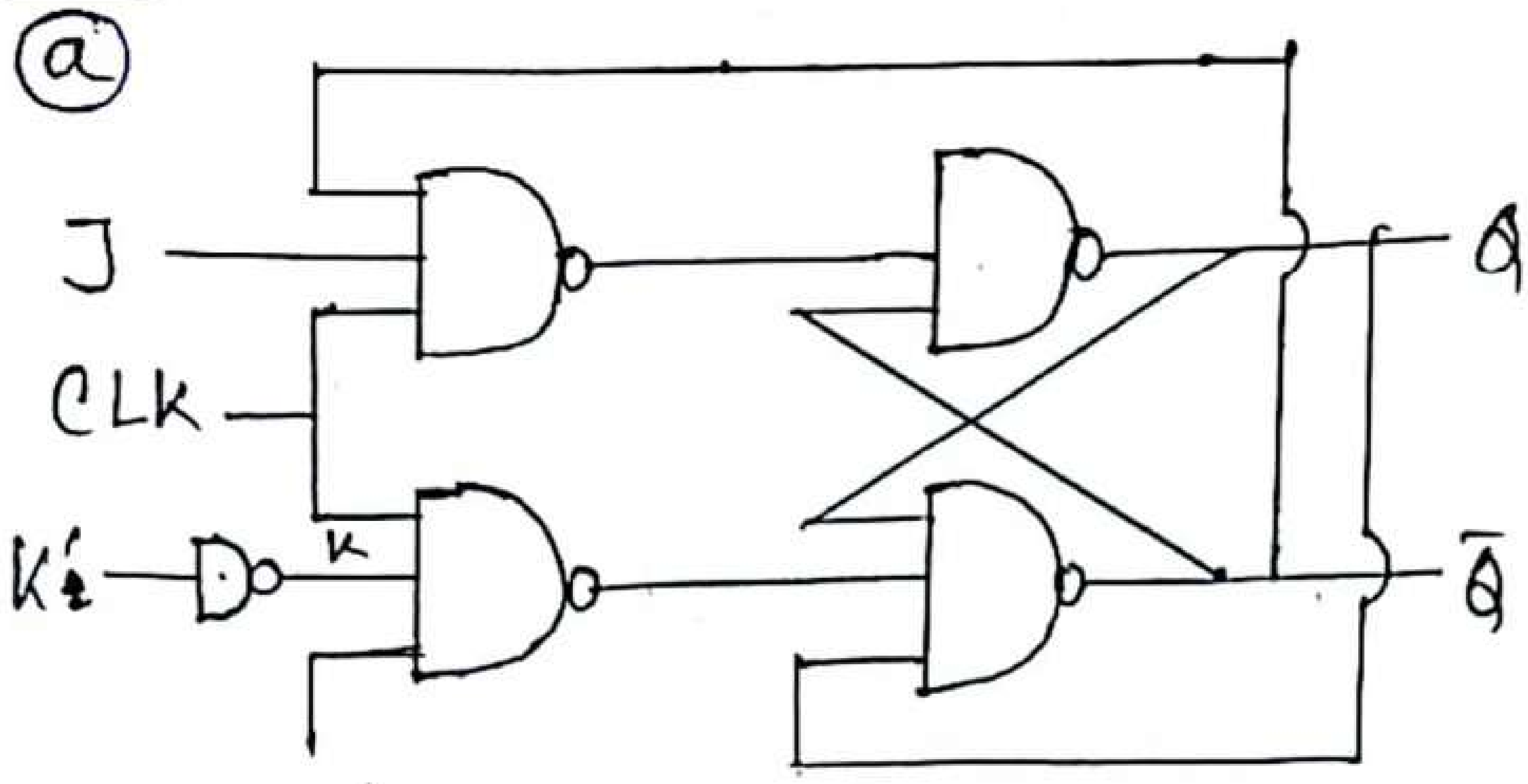
6.3

From Fig. 6.5 (a) we can't reduce one gate. If reduce one gate, then we can't get 'D' flip-flop.



6.4

(a)



Characteristic table

CLK	J	K'	Qn	Qn+1
0	x	x	x	Qn
1	0	0	0	0
2	0	0	1	0
3	0	1	0	0
4	0	1	1	1
5	1	0	0	1
6	1	0	1	1
7	1	1	0	1
8	1	1	1	0

(b) Characteristic equation :

$Q_{n+1} = \Sigma m(3,4,5,6)$

J \ K'Qn	00	01	11	10
0			1	
1	1	1		1

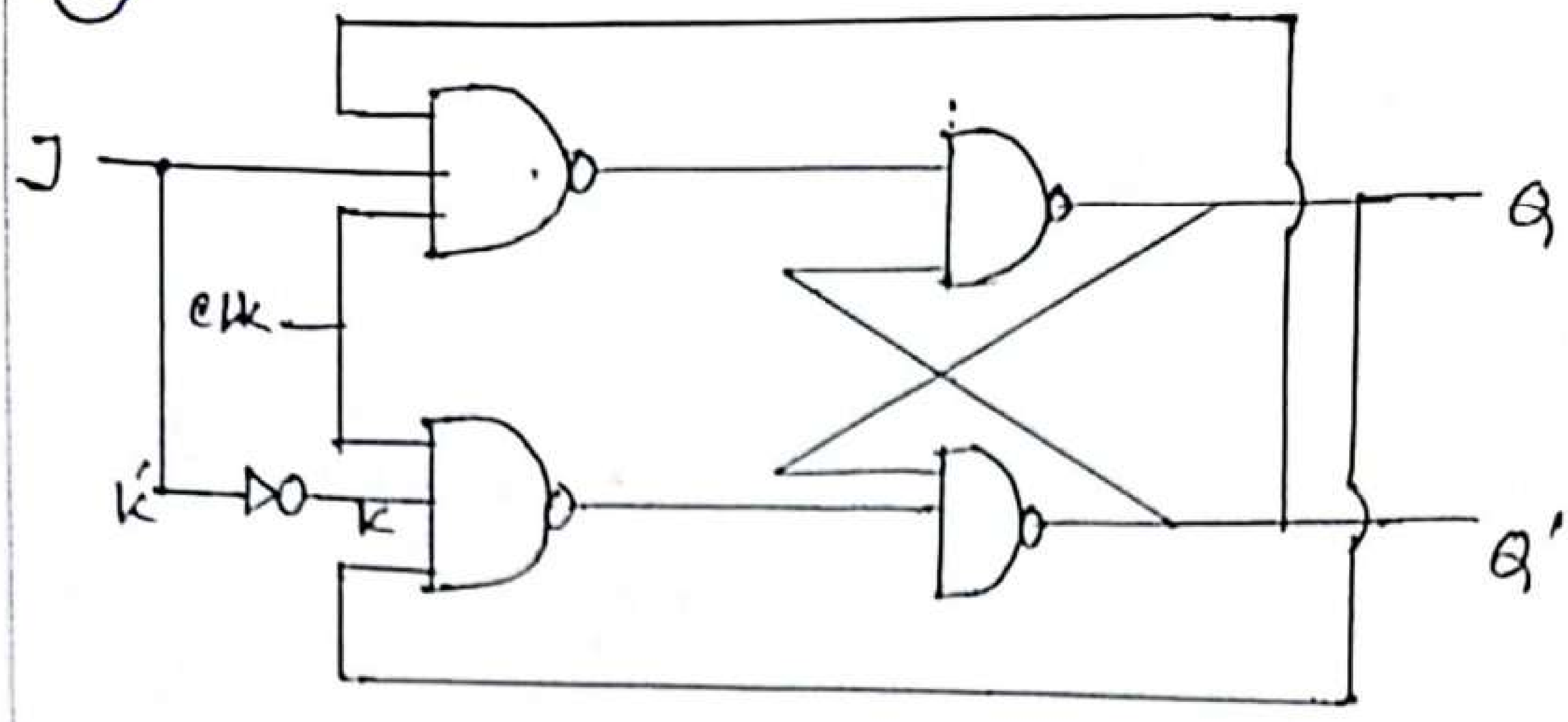
$Q_{n+1} = Jk + J\bar{Q}_n + J'KQ_n$

JK flip-flop

Truth table:

J	K	Qn+1
0	0	Qn
0	1	0
1	0	1
1	1	Qn'

(c)



CLK	J	Q_n	Q_{n+1}
0	x	x	Q_n
1	0	0	0
2	0	1	0
3	1	0	1
4	1	1	1

$Q_{n+1} = \Sigma m(2,3)$

$Q_{n+1} = J$

This is equal to D flip-flop charactaristic equation, so if we tyed both J & K' we get D flip-flop.

6.5

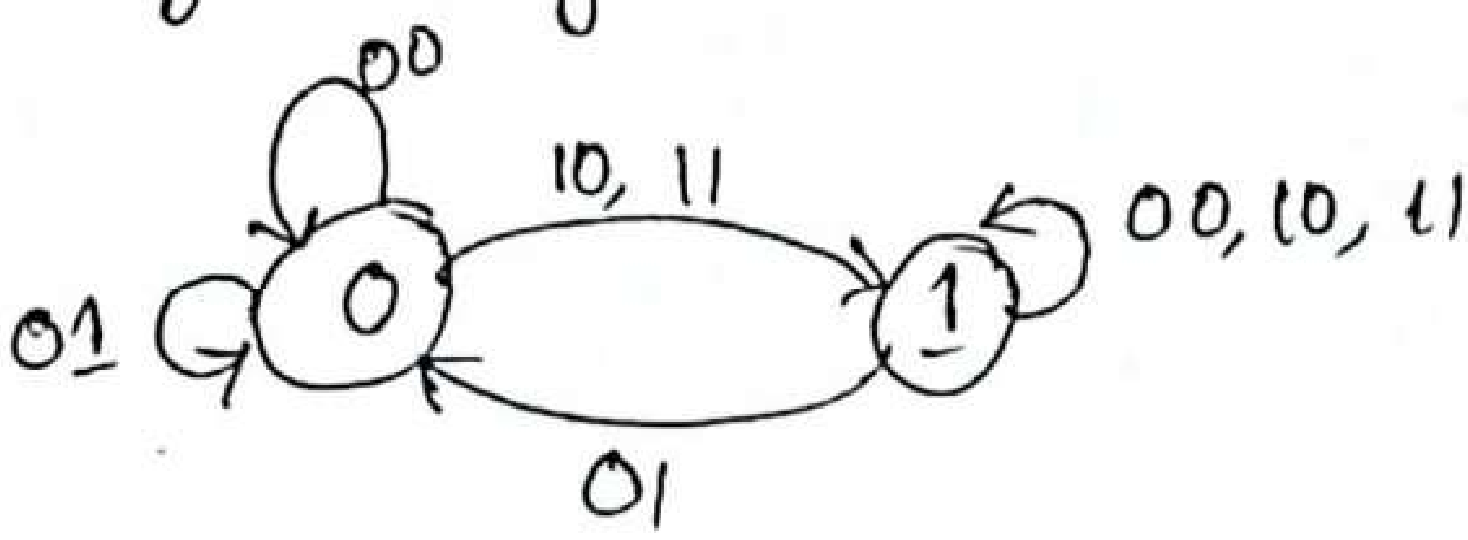
S	R	Q
0	0	0
0	1	0
1	0	1
1	1	1

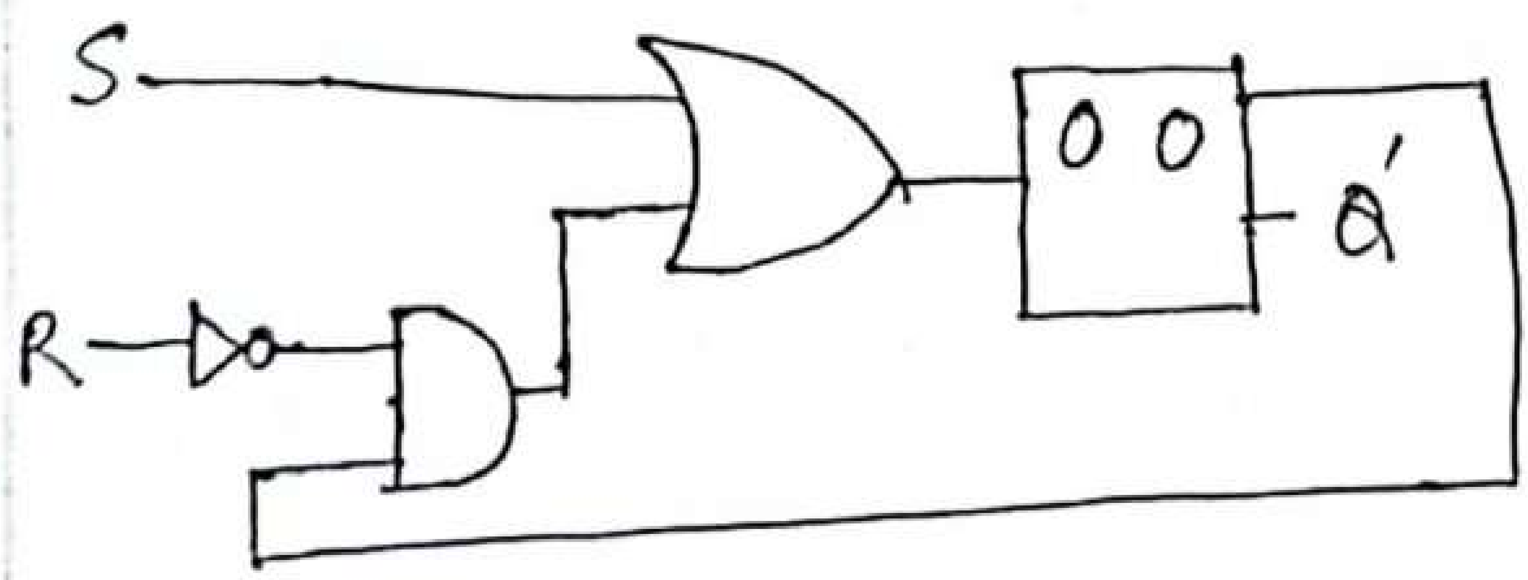
S	R	Q	Q'	D
0	0	0	0	0
0	1	0	0	0
1	0	0	1	1
1	1	0	1	1
0	0	1	1	1
0	1	1	0	0
1	0	1	1	1
1	1	1	1	1

Q'	R'Q'	R'Q	RQ	RQ'
S'		1		
S	1	1	1	1

$$Q' = S + R'Q$$

Logic Diagram :





Implementation with D-flip-flop

6.6

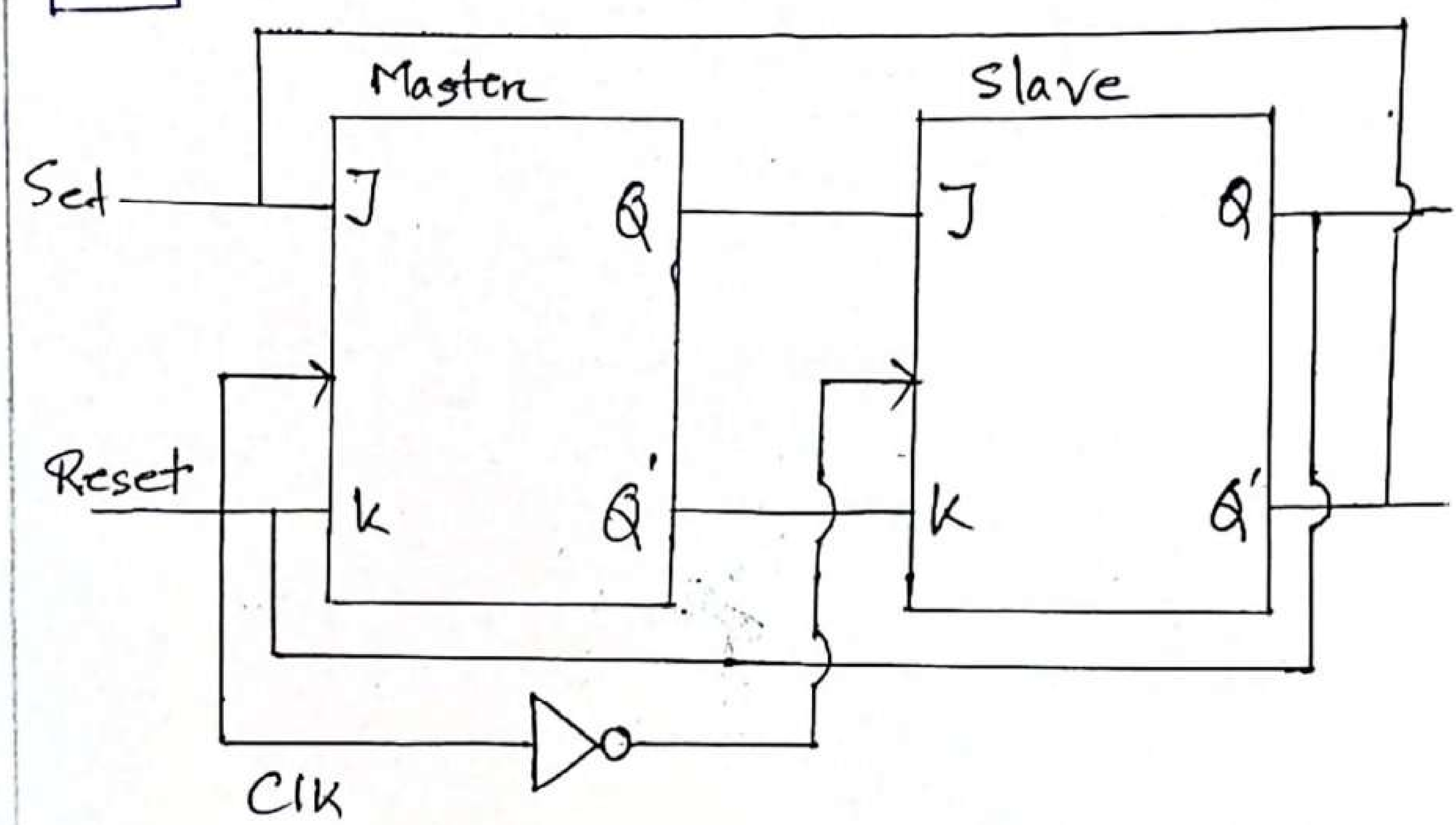
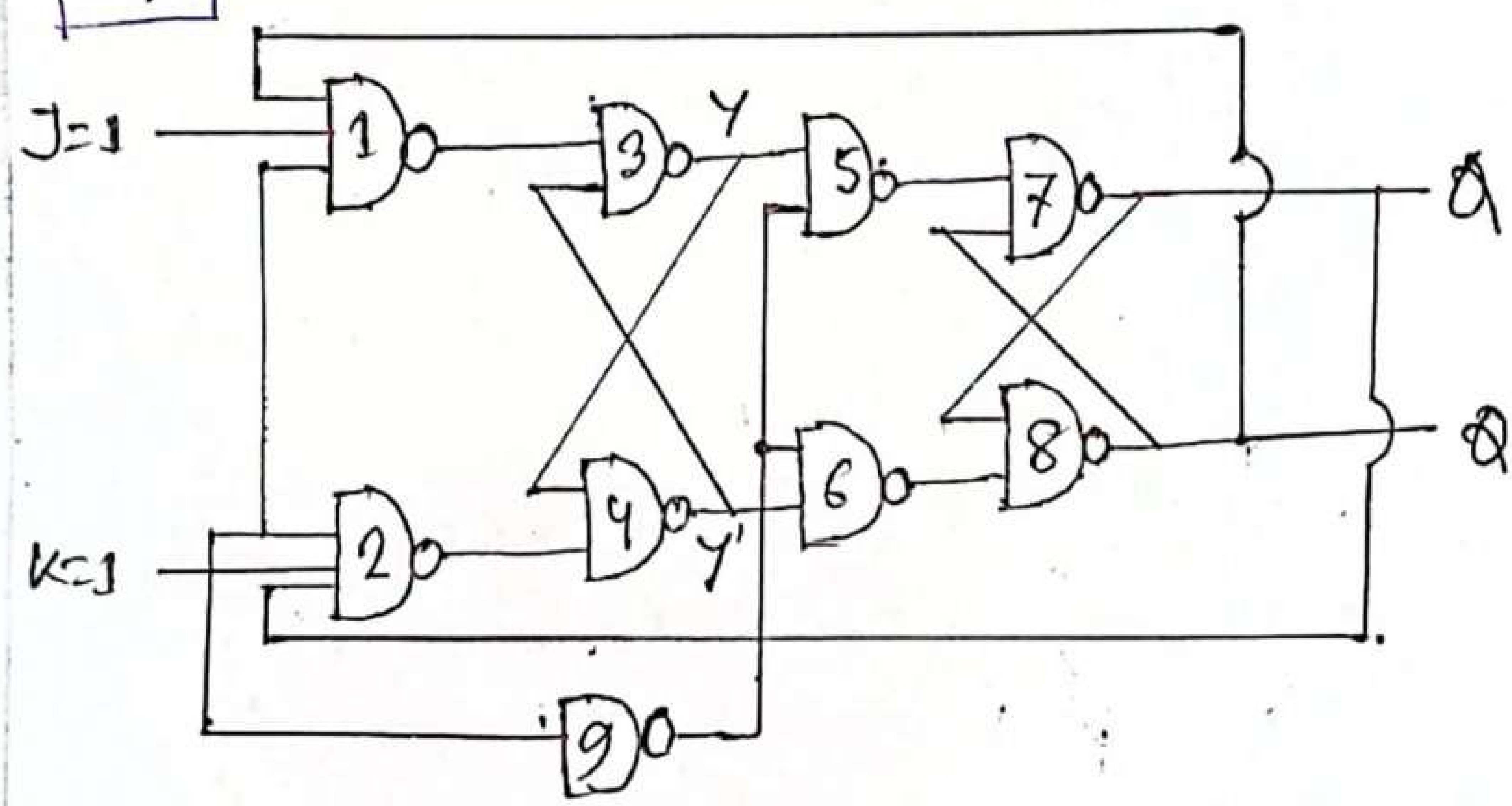


Fig : Master-Slave JK-flip-flop

6.7



ⓐ CP = 0, y = 0, Q = 0, J = K = 1

The above circuit is a master-slave flip-flop. Master output of JK flip-flop

~~C35~~

is fed to the input of the slave JK flip-flop. The output of the slave JK flip-flop is given as a feedback to the output of master JK flip-flop. The clock pulse is given to the master flip-flop and it is sent through a NOT gate ans thus invested passing it to the slave JK flip-flop.

Operation:

Master flip-flop is a positive level triggered. But due to the presence of the inverter in the clockline, the slave will respond to the negative level. When clock = 1 (Positive level) the master is active and the slave ~~it~~ is inactive whearas. When clock = 0, the slave is active and master is inactive.

So the given i/p values, slave this flip-flop is active because clock = 0

The output of slove Q = 0 because input of slave is J = 0, K = 1.

so the condition, the output of slave flip-flop $Q=0$ (it is a reseet condition)

the clock $=0$, so the master output is not changed.

The final output values of all are:

"Gate 1 $=0$, Gate 2 $=1$, Gate 3 $=0$, Gate 4 $=0$,

Gate 5 $=1$, Gate 6 $=0$, Gate 7 $=0$, Gate 8 $=1$,

Gate 9 $=1$"

(b) CP goes to 1 (Y should go to 1: Q remain at $=0$)

Now clock pulse $=1$, master flip flop is active $J=1$, $K=0$ thes at is set condition. I.e. the output of master is high (slave is inactive)

Gate 1 $=0$, Gate 2 $=1$, Gate 3 $(Y)=1$,

Gate 4 $(Y')=0$, Gate 5 $=1$, Gate 6 $=1$

Gate 7 $(Q)=0$, Gate 8 $(Q')=1$, Gate 9 $=1$.

(c) After CP goes to 0 and immedietly after that 5 goes to 0 (Q should go to 1)

Clock pulse $=0$, theree slave is active and master flip-flop is inactive. Therefore output of master is not changed.

output of all gives are:

Gate 1 = 1, Gate 2 = 1, Gate 3 (Y) = 1

Gate 4 = 0, Gate 5 = 0, Gate 6 = 1

Gate 7 (Q) = 1, Gate 8 (Q') = 0, Gate 9 = 0

(d)

After CP goes to 1 (Y should go to 0) Now master is active and slave is inactive. it's a reset condition. Therefore output of $y = 0$, $y' = 1$

Gate 1 = 1, Gate 2 = 0, Gate 3 (Y) = 0,

Gate 4 = 0, Gate 5 = 1, Gate 6 = 1

Gate 7 (Q) = 1, Gate 8 (Q') = 0, Gate 9 = 1

(e)

After CP goes back to 0 & immedictly after that K goes to 0 (Q should go to 0).

Now clock pulse = 0, slave is active. It is a reset condition at slave flip-flop, so the output of slave flip-flop is 0.

Gate 1 = 1, Gate 2 = 1, Gate 3 (Y) = 0

Gate 4 = 1, Gate 5 = 1, Gate 6 = 0

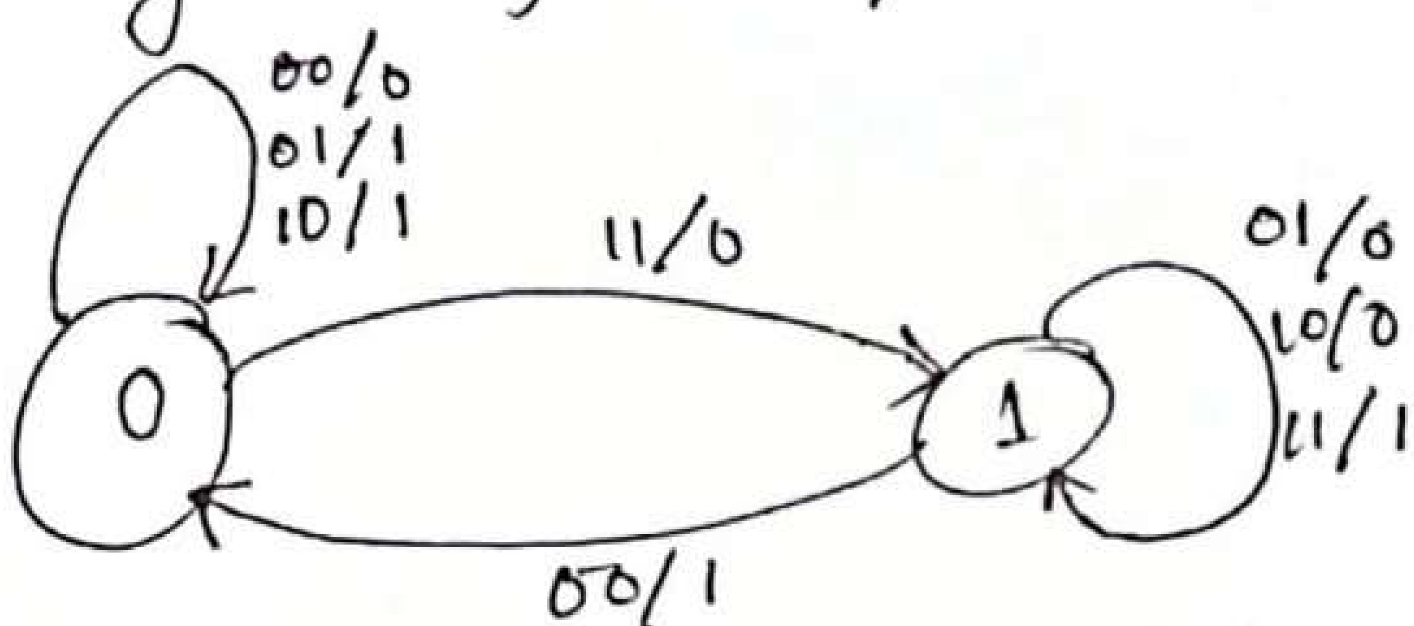
Gate 7 (Q) = 0, Gate 8 = 1, Gate 9 = 0

6.9

State table of the sequencial circuit

Present state, Q(t)	Input		Next state, Q (t+1)	Output
	x	y		S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

State diagram of a sequencial circuit:



It is a ripple carry adder because the carry output of the Full-adder is feed-back to the next addition as carry in.

6.11 (a)

Given equation:

$$A(t+1) = (CD' + C'D)x + (CD + C'D')x'$$

$$B(t+1) = A, \quad C(t+1) = B, \quad D(t+1) = C$$

It is given that $x = 1$ & $ABCD = 0001$

$$A(t+1) = C \oplus D$$

0001 → 1000 → 0100 → 0010 → 1001 → 1100 → 0110 → 1011 → 0101 → 1010 → 1101 → 1110 → 1111 → 0111 → 0011 → 0001

Here we simply calculate the XOR value between C & D and then we send it to the A.

So, in total we have 15 different states

P	Q	XOR
0	0	0
0	1	1
1	0	1
0	0	0

CS CamScanner

b) When, input $X = 0$

$A(t+1) = C \odot D \quad B(t+1) = A$

The truth table for XNOR gate is

P	Q	XNOR
0	0	1
0	1	0
1	0	0
1	1	1

0000 → 1000 → 1100 → 1110

1101 ← 1011 ← 0111 ←

→ 0110 → 0011 → 1001 → 1100

1101 ← 1011 ← 0111 ← 1110 ←

→ 0110 → 0011 → 1001 → 0100

0001 ← 0010 ← 0101 ← 1010 ←

→ 0000

In total, we have 25 different states.

6.27

BASIS FOR Comparison	Combinational circuit	Sequential circuit
Basic	The output Is discovered by the present state of the inputs.	Both the present input & past state output are used to identify the output
Storage Capability	Does not store data	Can store a small amount of data
Application	Used in adders, encoders, multiplexer etc	Flip-flop latches
Clock	Circuits do not rely on the clock	Clock is utilized for performing triggering functions
Feedback	No requirement of the feedback	Feedback is required

6.18 Excitation Table:

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

5.29

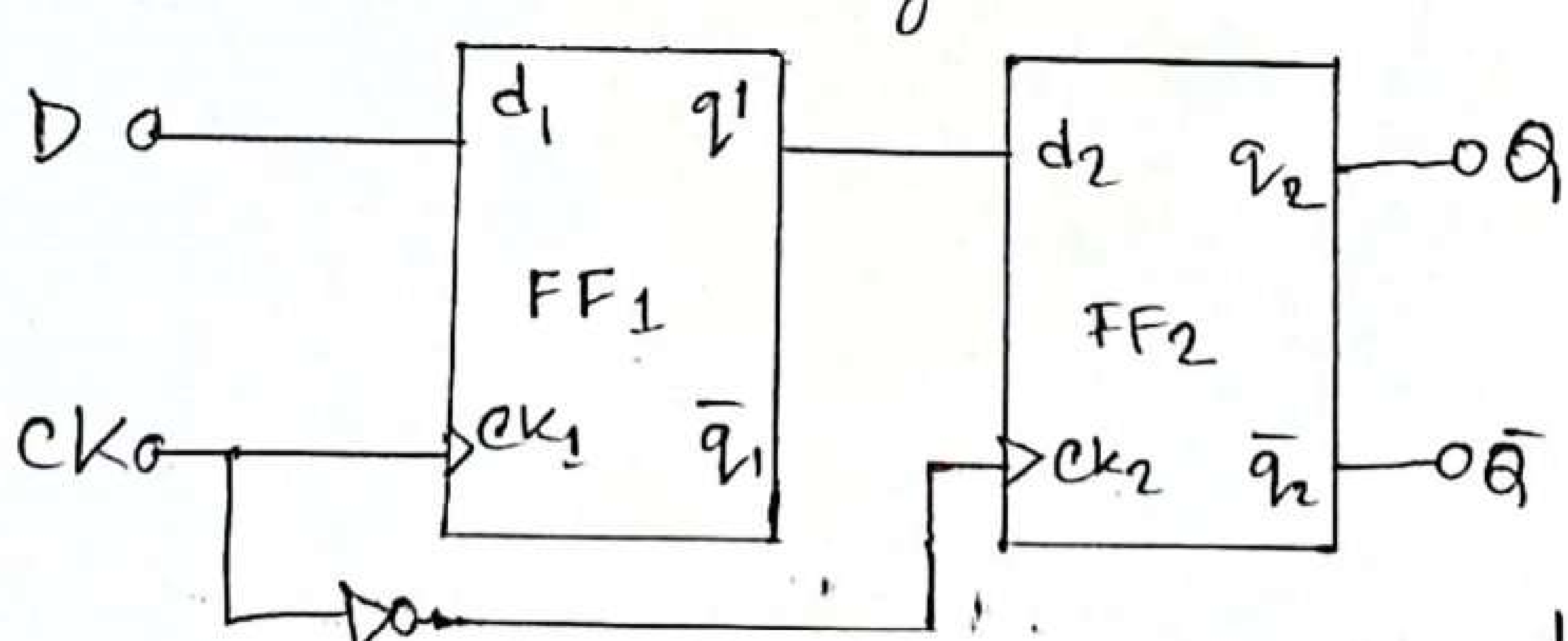
Basically, this type of flip-flop can be designed with two JK FFs by connecting in series. The connection of thes FFs one FF works as the master as well as a slave.

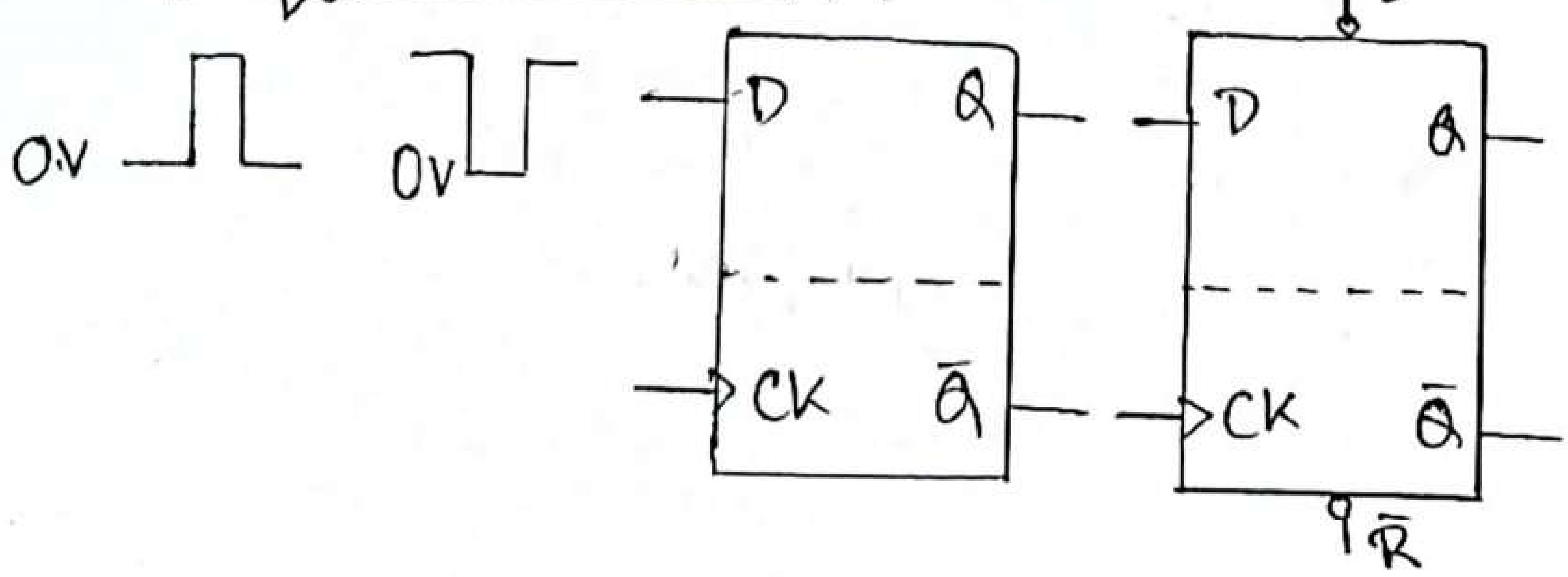
In this type of FF an inverter is also used addition to two FFs. The inverter connection can be done in such a way that where the invented clk pulse can be connected to the the slave FF. In other terms, if clk pulse is 0 for a master FF, then clk pulse will be 1 for a slave FF. Similarly, when clk pulse is 1 for master, then clk pulse will be 0 for slave FF.

Race around condition in JK flip-flop to for JK flip-flop, if $J=K=1$ an if $clk=1$ for a long period of time, then Q output will toggle as CLK is high, which makes the output of the flip-flop unstable. This problem is called race around condition in JK flip-flop. This problem can be avoided by ensuring that the clock input is at logic 'I' only for a very short time. This introduced the concept of Master Slave JK flip-flop.

The D type Master-Slave Flip-flop :

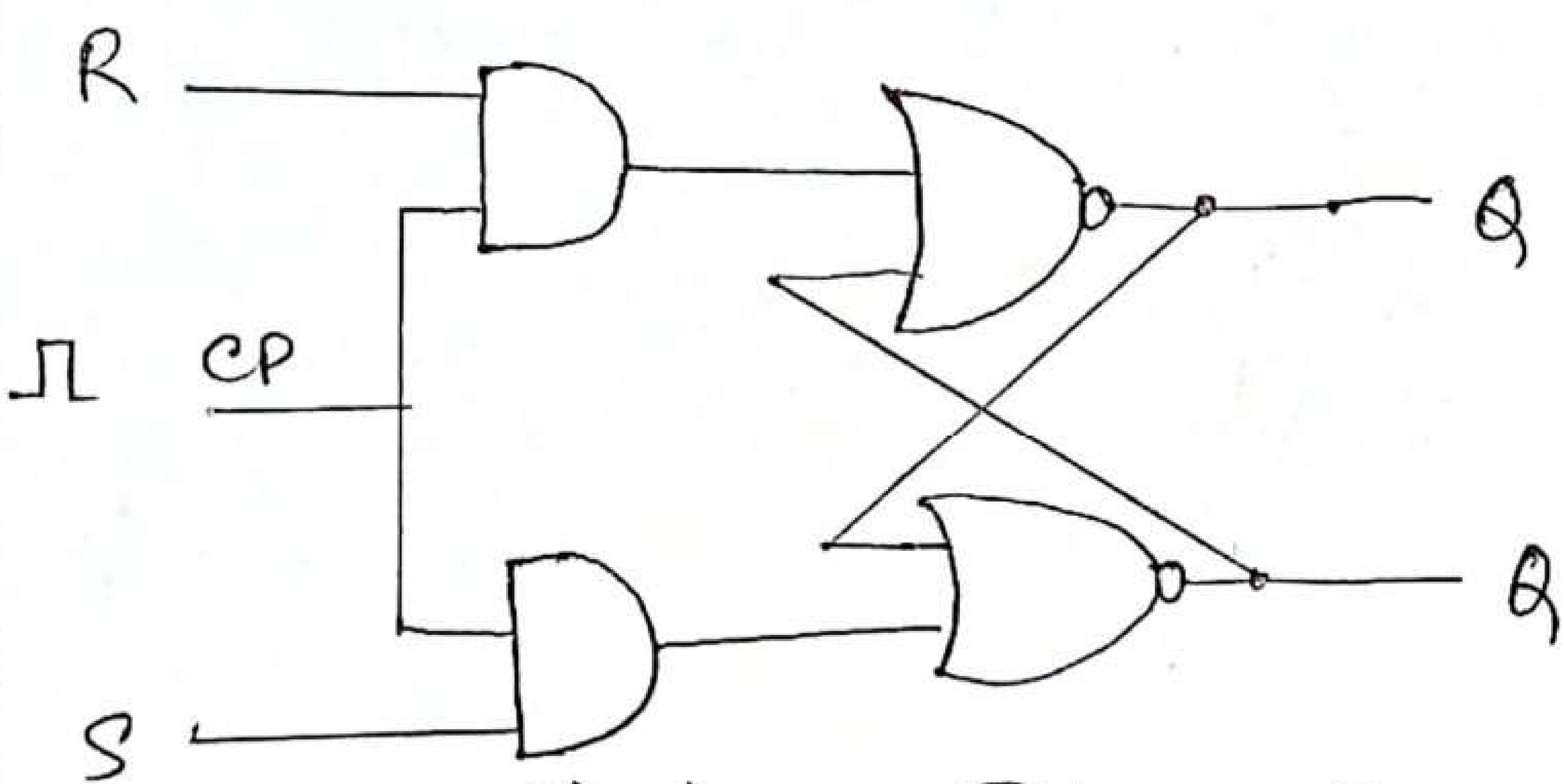
A further version of the D type flip-flop is shown in below figure; where two D type flip-flop are incorporated in a single device. This is the D type master slave flip flop. Circuit symbols for the master-slave device are very similar to those for edge-triggered flip-flop, but are now divided into two sections by a dotted line,





6.28

The problems with SR flip-flops using NOR and NAND gate is invalid state. This problem can be overcome by using a bistable SR flip-flop that can change outputs when certain invalid states are met, regardless of the condition of either the set or the Reset inputs. For this a clocked SR flip flop is designed by adding two AND gates to a basic NOR gate flip-flop. The circuit diagram and truth table is shown below.

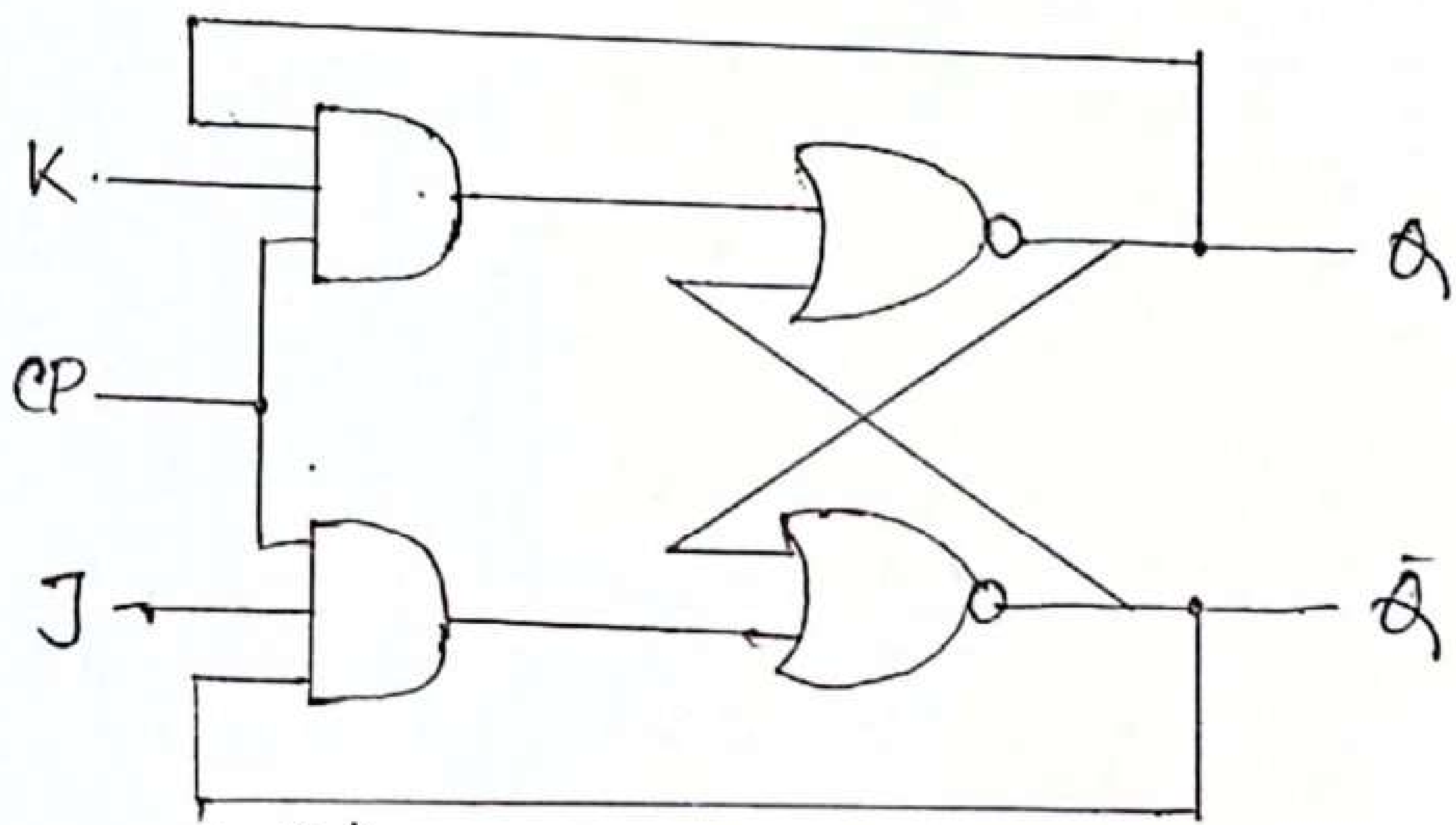


(a) Logic Diagram

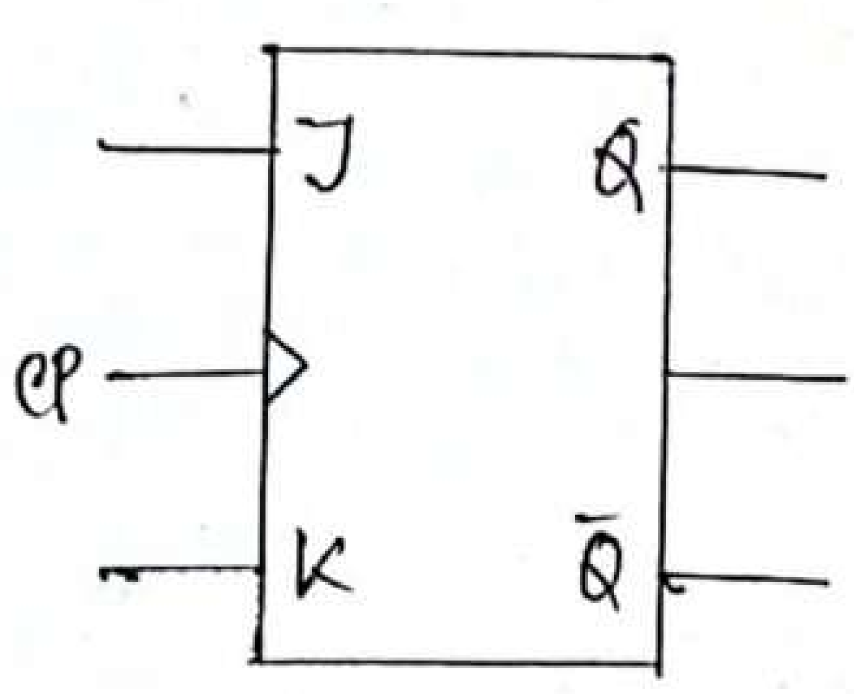
Q	S	R	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	indeterminate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	indeterminate

b) Truth table

A clock pulse is given to the inputs of the AND gate. When the value of the clock pulse is '0' the outputs of the both the AND gates remain '0'. As soon as a pulse is given the value of CP turns '1'. This makes the values at S and R to pass through the NOR gate flip-flop. But when the values of the both S an R turns '1' the High value of CP causes both of them to turn to '0' for a short moment.



(a) Logic Diagram



(b) Graphical symbol

Q	J	K	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(c) Truth table

Clocked JK flip-flop

A JK flip-flop can also be defined as a modification of the SR flip flop. The only difference is that the intermediate state is more refined and precise than that of a SR flip-flop.

The behaviour of inputs J and K is same as the S and R inputs of the SR flip-flop. The letter J stands for SET and the letter K stands for CLEAR its same to SR flip-flop. The letter J and K stands for SET & CLEAR respectively.

When both the inputs J & K have a HIGH state, the flip-flop swith to the compliment state. So, for a value of Q = 1, it switches to Q = 0 and for a value of Q = 0, it switches to Q = 1.

The circuits include two 3-input AND gates. The output Q of the flip-flop is returned back as a feedback to the end input of the AND along with other inputs like k and clock pulse[cp] .. So, if the value of CP is '1' the flip flop gets a CLEAR signal and with the condition that the value of Q was earlier 1. Similarly output 'Q'' of the flip flop is given as a feedback to the input of the AND along with other inputs like J and clock pulse [CP]. So, the output becomes SET when the value of CP is 1 only if the value of Q' was earlier 1.

The output may be repeated in transitions once they have been complimented for J=K=1 because of the feedback connection in the JK flip-flop. This can be avoided by setting a time duration lesser than the propagation delay through the flip-flop. The restriction on the pulse width can be eliminated with a master-slave.

6.30

Inputs			Output
D	Previous state Q_n	Next state Q_{n+1}	T
0	0	0	0
1	0	1	1
0	1	0	1
1	1	1	0

Fig: Excitation table

For output D

$D \backslash Q_n$	0	1
0	0	1
1	1	0

$\rightarrow \bar{D} Q_n$

$\rightarrow D \bar{Q}_n$

$T = DQ_n + DQ_n$

$\therefore T = D \oplus Q_n$

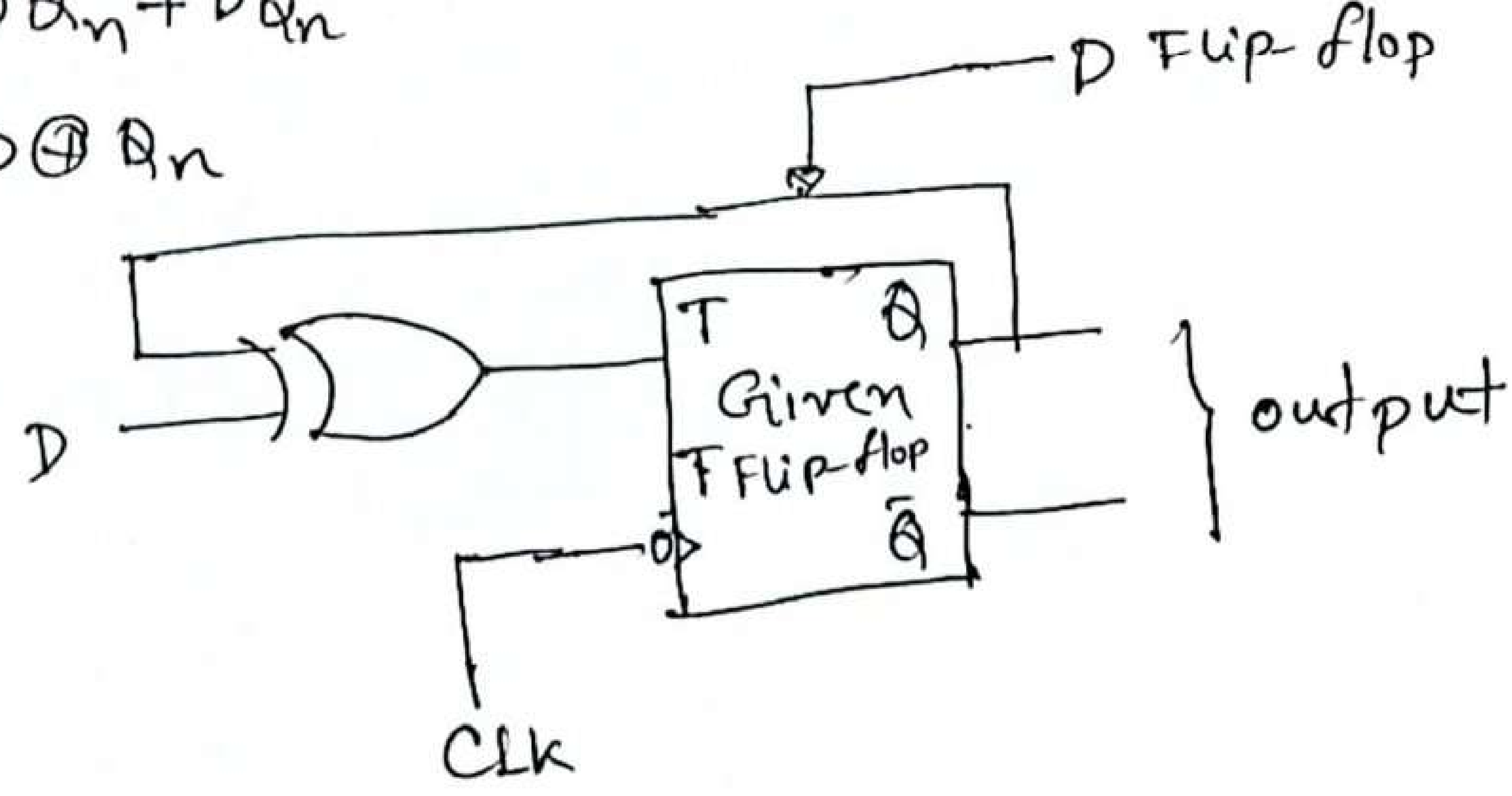


Fig: Conversion of T Flip-flop to D Flip-flop

6.36

A level triggered flip-flop will register a "one" asserted at an input at any time while the clock input in the ~~rep~~ 'recepted' state, even if that 'one' deasserted prior to the end of the receptive period. At the places where I worked most actively at digital design, we called these 'ones catchers'.

An edge-triggered flop basically looks at the driving inputs only at the precise instant the clock signal changes.

In most cases, edge triggered flops are easier to use because they are not sensitive to transient conditions that might arise while the inputs 'settle down' from the most recent state change. With level-triggered flops, all transients must settle before the clock signal moves to the receptive state.